AP CALCULUS AB

Unit 1 • Worksheet 1-2A • Answer key

Worksheet 1-2A • Limits to a Point of a Function

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is $\lim_{x \to 2} \left(3x^2 - x + 1\right)$?

(A) 11

(B) 13

(C) 9

(D) It does not exist.

Solution

Answer: (A). Plug in: $12 - 2 + 1 = 11$.

Question 2

NO CALCULATOR

What is $\lim_{x \to -1} \left(x^3 + 2x\right)$?

(A) 3

(B) -3

(C) 1

(D) -1

Solution

Answer: (B). $-1 - 2 = -3$.

Question 3

NO CALCULATOR

What is $\lim_{x \to 4} 7$?

(A) 4

(B) 0

(C) 7

(D) 28

Solution

Answer: (C). A constant function: the limit is the constant.

Question 4

NO CALCULATOR

Let $p(x) = \begin{cases} x^2 - 3, & x < 2 \\ 2x - 3, & x \geq 2. \end{cases}$ What is $\lim_{x \to 2} p(x)$?

(A) It does not exist.
(B) 1
(C) 4
(D) -1

Solution
Answer: (B). Left: $4 - 3 = 1$. Right: $4 - 3 = 1$. The limit is 1.

Question 5

NO CALCULATOR

Let $q(x) = \begin{cases} 5 - x, & x \leq 3 \\ x^2 - 6, & x > 3. \end{cases}$ What is $\lim_{x \to 3} q(x)$?

(A) 2
(B) 3
(C) It does not exist.
(D) $\frac{5}{2}$

Solution
Answer: (C). Left: $5 - 3 = 2$. Right: $9 - 6 = 3$. Different: DNE.

Question 6

NO CALCULATOR

For the same q, what is $q(3)$?

(A) 2
(B) 3
(C) It does not exist.
(D) 5

Solution
Answer: (A). $x = 3$ belongs to the top piece: $5 - 3 = 2$.

Question 7

NO CALCULATOR

What is $\lim_{x \to 5} \dfrac{x^2 - 25}{x - 5}$?

(A) 0
(B) 10
(C) 5
(D) It does not exist.

Solution
Answer: (B). $\frac{0}{0}$. $\frac{(x-5)(x+5)}{x-5} = x + 5 \to 10$.

Question 8

NO CALCULATOR

What is $\lim_{x\to -2} \dfrac{x^2+5x+6}{x^2-4}$?

(A) $\frac{1}{4}$
(B) 0
(C) It does not exist.
(D) $-\frac{1}{4}$

Solution
Answer: (D). $\frac{(x+2)(x+3)}{(x+2)(x-2)} = \frac{x+3}{x-2} \to \frac{1}{-4}$.

Question 9

NO CALCULATOR

What is $\lim_{x\to 0} \dfrac{2x^2+6x}{x^2-3x}$?

(A) -2
(B) 2
(C) 0
(D) It does not exist.

Solution
Answer: (A). $\frac{2x(x+3)}{x(x-3)} = \frac{2(x+3)}{x-3} \to \frac{6}{-3} = -2$.

Question 10 (Free response)

NO CALCULATOR

(a) Find $\lim_{x\to 3} \dfrac{x^2-x-6}{x^2-9}$. Show the three steps.

(b) Let $h(x) = \begin{cases} x+4, & x<1 \\ x^2+4, & x\geq 1. \end{cases}$ Find $\lim_{x\to 1^-} h(x)$, $\lim_{x\to 1^+} h(x)$, $\lim_{x\to 1} h(x)$ and $h(1)$.

(c) A classmate says: "Plugging in gives $\frac{0}{0}$, so the limit is 0." Explain the mistake.

Solution
(a) $\frac{0}{0}$. Factor: $\frac{(x-3)(x+2)}{(x-3)(x+3)}$; cancel: $\frac{x+2}{x+3}$; plug in: $\frac{5}{6}$.
(b) 5, 5, 5, 5.
(c) $\frac{0}{0}$ is an indeterminate form, not a value. We must factor, cancel and plug in again.